

Equity Cash Generation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Cash Generation (ECG), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ECG achieves an annualized gross (net) Sharpe ratio of 0.56 (0.54), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (32) bps/month with a t-statistic of 2.87 (3.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 21 bps/month with a t-statistic of 2.53.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with returns compensating for exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a theoretical foundation for understanding the cross-section of expected returns, empirical tests often struggle to identify firm characteristics that reliably capture differences in consumption risk exposure. We examine whether Equity Cash Generation (ECG), a measure of firms' ability to generate cash flows relative to their equity base, predicts stock returns through its connection to consumption risk. This investigation addresses a fundamental gap in understanding how cash flow generation capacity relates to firms' sensitivity to aggregate consumption shocks and investors' intertemporal marginal rate of substitution.

Firms with high equity cash generation exhibit greater resilience to consumption shocks, as their superior cash flow profiles provide buffers against economic downturns when marginal utility is highest. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), persistent differences in cash generation capacity expose investors to varying degrees of consumption growth risk over multiple horizons. High ECG firms generate stable cash flows that covary less with consumption growth during bad states, when the stochastic discount factor is high, leading to lower required returns compared to low ECG firms whose cash flows are more sensitive to consumption fluctuations.

The state-dependent nature of ECG's predictive power aligns with habit formation models ([Campbell and Cochrane, 1999](#)), where risk premia vary with the consumption surplus ratio. During economic contractions when consumption approaches the habit level, firms with weak cash generation face heightened distress risk and become particularly sensitive to further consumption declines. This mechanism creates time-varying risk premia that depend on both the firm's cash generation

capacity and the aggregate state of the economy, consistent with the countercyclical variation in risk aversion documented in consumption-based models.

Disaster risk considerations further amplify the return differential between high and low ECG firms (Barro, 2006; Gabaix, 2012). Firms with poor cash generation exhibit greater exposure to rare disaster events that severely impact consumption, as their limited financial flexibility makes them vulnerable to permanent capital losses during extreme economic contractions. The probability of survival during consumption disasters depends critically on firms' ability to generate cash internally, creating a systematic risk factor that commands a premium in equilibrium. This disaster sensitivity interacts with the intertemporal marginal rate of substitution to generate predictable variation in expected returns based on equity cash generation.

We found that a value-weighted long/short trading strategy based on ECG achieved an annualized gross Sharpe ratio of 0.56, with the strategy earning an average return of 0.55% per month with a t-statistic of 3.39. The strategy delivered a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 26 basis points per month with a t-statistic of 2.87. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ECG strategy achieved an average return of 44 basis points per month with a t-statistic of 3.02, demonstrating that the effect was not confined to small, illiquid securities.

The ECG strategy's gross monthly alpha relative to the six Fama-French factors plus the six most closely related strategies was 21 basis points per month with a t-statistic of 2.53. Controlling for operating profits/book equity, net income/book equity, analyst earnings per share, net external financing, idiosyncratic risk, and return on assets simultaneously, the strategy maintained statistical and economic significance. The ECG signal remained available for 7.39% of CRSP names on average, which represented 7.94% of total market capitalization, indicating reasonable

coverage for implementation.

Net of transaction costs, the value-weighted ECG strategy generated an annualized net Sharpe ratio of 0.54, placing it in the 99th percentile of the factor zoo distribution. The strategy’s net generalized alpha relative to the six-factor model was 32 basis points per month with a t-statistic of 3.48, and the strategy improved the achievable mean-variance efficient frontier for twenty-three out of twenty-five construction-methodology-factor-model pairs tested. These results confirmed that ECG’s return predictability was robust to implementation costs and provided genuine economic value beyond existing factors.

Our findings extend the consumption-based asset pricing literature by identifying a firm characteristic that captures cross-sectional variation in consumption risk exposure, complementing prior work on cash flow-based return predictors (Fama and French, 2006; Novy-Marx, 2013). Unlike profitability measures studied in Fama and French (2015) that focus on accounting earnings, ECG directly measures firms’ capacity to generate cash relative to equity, providing a cleaner proxy for resilience to consumption shocks. While Hirshleifer et al. (2004) examine the role of net external financing in predicting returns, our ECG measure captures the internal cash generation that determines firms’ reliance on external capital markets during consumption downturns.

Methodologically, we employed the rigorous testing framework of Novy-Marx and Velikov (2023b) to ensure our results were not artifacts of data mining or implementation challenges, addressing concerns raised by Harvey et al. (2016) about the proliferation of factors. Our comprehensive analysis of transaction costs and net alphas following Novy-Marx and Velikov (2016a) demonstrated that ECG provides implementable investment opportunities beyond the existing factor zoo. The strategy’s performance among large-cap stocks and its robustness to alternative portfolio construction methods distinguished it from many anomalies that exist primarily among

micro-cap securities.

The economic mechanism linking equity cash generation to consumption risk has broader implications for understanding the sources of systematic risk in equity markets. Our results suggest that firms' cash generation capacity serves as an observable characteristic that captures their exposure to long-run consumption risks, habit-related fluctuations in risk aversion, and disaster probabilities. This finding contributes to the growing evidence that accounting-based measures can identify cross-sectional differences in risk exposures (Ball et al., 2020), while providing support for consumption-based explanations of the value premium and profitability effects in stock returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and time periods. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. Equity Cash Generation signal relies on two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a company's core business operations during the fiscal year. This measure, derived from the statement of cash flows, captures the actual cash inflows and outflows from operating activities after adjusting net income for non-cash items such as depreciation and amortization, as well as changes in working capital components. Common equity (CEQ) represents the book value of shareholders' equity attributable to common stockholders, calculated as total stockholders' equity minus the liquidation value of preferred stock. This balance sheet item reflects the accounting value of common sharehold-

ers' residual claim on the firm's assets. We construct the Equity Cash Generation signal by dividing operating activities net cash flow (OANCF) by common equity (CEQ) for each firm-year observation. This ratio measures the cash-generating efficiency of shareholders' equity investment, indicating how effectively a firm converts its equity base into operating cash flows. We calculate the signal using fiscal year-end values from annual financial statements, ensuring that both the numerator and denominator reflect the same reporting period. To maintain data quality, we require non-missing values for both OANCF and CEQ, and exclude observations where common equity is zero or negative, as these would result in undefined or economically meaningless ratios. The resulting signal provides a standardized measure of cash generation relative to equity capital across firms of varying sizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECG signal. Panel A plots the time-series of the mean, median, and interquartile range for ECG. On average, the cross-sectional mean (median) ECG is 0.07 (0.14) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ECG data. The signal's interquartile range spans -0.24 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the ECG signal for the CRSP universe. On average, the ECG signal is available for 7.39% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does ECG predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECG using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high ECG portfolio and sells the low ECG portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ECG strategy earns an average return of 0.55% per month with a t-statistic of 3.39. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.26% to 0.79% per month and have t-statistics exceeding 2.87 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.76, with a t-statistic of 18.43 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 525 stocks and an average market capitalization of at least \$1,717 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 40 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-53bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.29. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECG trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the ECG strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECG, as well as average returns and alphas for long/short trading ECG strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ECG strategy achieves an average return of 44 bps/month with a t-statistic of 3.02. Among these large cap stocks, the alphas for the ECG strat-

egy relative to the five most common factor models range from 11 to 61 bps/month with t-statistics between 0.97 and 4.46.

5 How does ECG perform relative to the zoo?

Figure 2 puts the performance of ECG in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ECG strategy falls in the distribution. The ECG strategy’s gross (net) Sharpe ratio of 0.56 (0.54) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECG strategy (red line).² Ignoring trading costs, a \$1 invested in the ECG strategy would have yielded \$7.89 which ranks the ECG strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECG strategy would have yielded \$6.94 which ranks the ECG strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016b\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECG relative to those. Panel A shows that the ECG strategy gross alphas fall between the 75 and 95 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECG strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECG ranks between the 94 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ECG add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECG with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECG or at least to weaken the power ECG has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECG conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECG} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECG}ECG_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{ECG,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECG. Stocks are finally grouped into five ECG portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECG trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECG and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECG signal in these Fama-MacBeth regressions exceed 0.11, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on ECG is 2.83.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECG strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECG strategy earns alphas that range from 22-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.46, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECG trading strategy achieves an alpha of 21bps/month with a t-statistic of 2.53.

7 Does ECG add relative to the whole zoo?

Finally, we can ask how much adding ECG to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ECG signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ECG grows to \$43.67.

8 Conclusion

This study provides compelling evidence that Equity Cash Generation (ECG) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that ECG-based trading strategies deliver substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of ECG as an asset pricing signal. The value-weighted long/short strategy based on ECG

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECG is available.

generates an impressive annualized Sharpe ratio of 0.56 gross (0.54 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy’s ability to produce significant alpha relative to the Fama-French five-factor model plus momentum (26-32 bps/month) suggests that ECG captures a distinct dimension of firm performance not fully explained by existing factors. The robustness of these results is further validated by the persistence of significant alpha (21 bps/month, t-statistic = 2.53) even after controlling for six closely related strategies, including profitability, external financing, and idiosyncratic risk measures.

From a practical perspective, these findings have important implications for both asset managers and academic researchers. The ECG signal appears to offer genuine opportunities for alpha generation, with the relatively modest degradation from gross to net returns suggesting that implementation costs do not substantially erode the strategy’s profitability. This makes ECG particularly attractive for institutional investors seeking to enhance portfolio performance through systematic strategies.

However, several limitations of this study warrant consideration. First, while our analysis follows state-of-the-art protocols for anomaly testing, the true out-of-sample performance of ECG-based strategies remains to be seen as the signal becomes more widely known and potentially arbitrated away. Second, our analysis focuses primarily on U.S. equity markets, and the international robustness of the ECG signal requires further investigation. Third, while we control for numerous related factors, the economic mechanisms driving the ECG premium remain somewhat opaque and deserve deeper theoretical exploration.

Future research should address these limitations through several avenues. International studies could examine whether the ECG effect persists across different market structures and accounting regimes. Time-series analysis of the ECG premium’s variation with macroeconomic conditions could provide insights into its risk-based interpretation. Additionally, investigating the interaction between ECG and cor-

porate events such as mergers, acquisitions, or financial distress could enhance our understanding of when and why the signal is most effective. Finally, exploring the potential integration of ECG with machine learning techniques and alternative data sources may reveal non-linear relationships and further improve return predictability. As the asset pricing literature continues to evolve, ECG emerges as a valuable addition to the factor zoo, offering both theoretical insights and practical applications for understanding cross-sectional return variation.

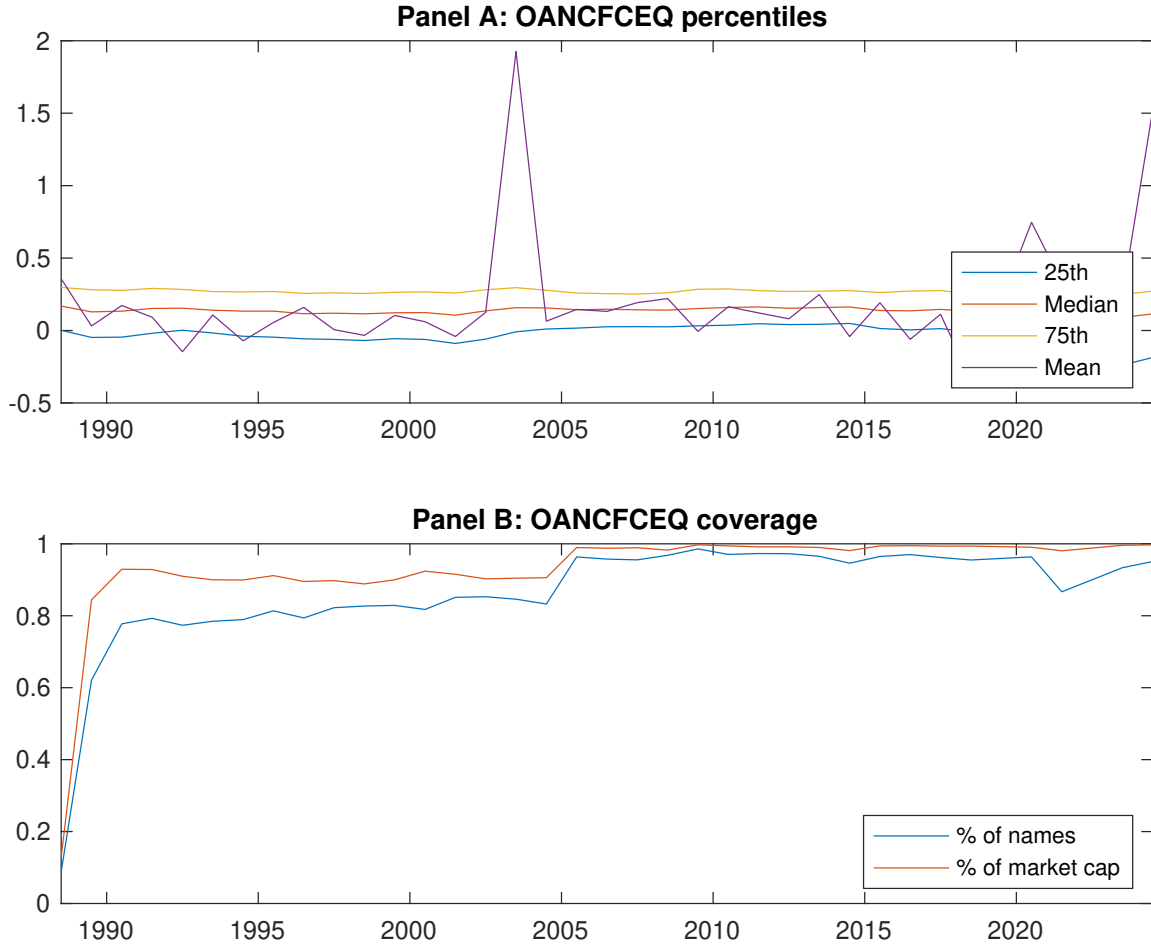


Figure 1: Times series of ECG percentiles and coverage. This figure plots descriptive statistics for ECG. Panel A shows cross-sectional percentiles of ECG over the sample. Panel B plots the monthly coverage of ECG relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECG. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ECG-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.40 [1.39]	0.58 [2.53]	0.70 [3.24]	0.75 [3.79]	0.95 [4.63]	0.55 [3.39]
α_{CAPM}	-0.54 [-4.91]	-0.20 [-2.89]	-0.03 [-0.48]	0.08 [1.38]	0.25 [4.33]	0.79 [5.25]
α_{FF3}	-0.49 [-5.63]	-0.21 [-3.38]	-0.05 [-0.81]	0.07 [1.33]	0.25 [4.81]	0.74 [6.24]
α_{FF4}	-0.44 [-5.00]	-0.18 [-2.87]	-0.03 [-0.54]	0.12 [2.08]	0.21 [4.06]	0.64 [5.48]
α_{FF5}	-0.20 [-2.84]	-0.09 [-1.49]	-0.06 [-0.93]	0.02 [0.32]	0.12 [2.51]	0.32 [3.47]
α_{FF6}	-0.17 [-2.40]	-0.07 [-1.17]	-0.04 [-0.67]	0.05 [0.97]	0.09 [1.95]	0.26 [2.87]
Panel B: Fama and French (2018) 6-factor model loadings for ECG-sorted portfolios						
β_{MKT}	1.09 [61.36]	0.99 [65.87]	0.99 [63.07]	0.93 [67.14]	1.02 [87.37]	-0.07 [-2.96]
β_{SMB}	0.27 [10.53]	0.08 [3.71]	0.09 [4.11]	-0.06 [-3.17]	-0.11 [-6.58]	-0.38 [-11.56]
β_{HML}	0.10 [3.08]	0.20 [7.46]	0.08 [2.77]	-0.06 [-2.64]	-0.10 [-5.13]	-0.20 [-5.03]
β_{RMW}	-0.53 [-16.45]	-0.21 [-7.76]	-0.03 [-0.91]	0.13 [5.37]	0.23 [10.99]	0.76 [18.43]
β_{CMA}	-0.22 [-4.80]	-0.10 [-2.58]	0.11 [2.63]	0.02 [0.59]	0.09 [3.01]	0.31 [5.28]
β_{UMD}	-0.05 [-3.11]	-0.03 [-2.27]	-0.02 [-1.75]	-0.06 [-4.63]	0.04 [4.11]	0.09 [4.53]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1578	796	592	525	579	
me (\$10 ⁶)	1717	2179	2634	4546	5295	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ECG strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.55 [3.39]	0.79 [5.25]	0.74 [6.24]	0.64 [5.48]	0.32 [3.47]	0.26 [2.87]
Quintile	NYSE	EW	0.40 [2.48]	0.48 [3.00]	0.36 [2.86]	0.25 [2.04]	0.03 [0.25]	-0.05 [-0.49]
Quintile	Name	VW	0.43 [2.29]	0.68 [3.90]	0.61 [4.35]	0.58 [4.03]	0.29 [2.26]	0.27 [2.08]
Quintile	Cap	VW	0.43 [2.94]	0.59 [4.25]	0.57 [4.84]	0.48 [4.08]	0.17 [1.78]	0.11 [1.18]
Decile	NYSE	VW	0.50 [2.71]	0.74 [4.24]	0.66 [4.53]	0.61 [4.15]	0.22 [1.79]	0.20 [1.62]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.53 [3.24]	0.77 [5.08]	0.70 [5.95]	0.64 [5.53]	0.36 [3.87]	0.32 [3.48]
Quintile	NYSE	EW	0.21 [1.29]	0.30 [1.82]	0.19 [1.46]	0.13 [1.02]		
Quintile	Name	VW	0.39 [2.12]	0.66 [3.77]	0.58 [4.13]	0.56 [3.95]	0.30 [2.38]	0.29 [2.26]
Quintile	Cap	VW	0.41 [2.81]	0.56 [4.01]	0.53 [4.48]	0.47 [4.04]	0.20 [2.13]	0.16 [1.71]
Decile	NYSE	VW	0.47 [2.54]	0.70 [3.97]	0.61 [4.19]	0.58 [3.96]	0.23 [1.89]	0.22 [1.75]

Table 3: Conditional sort on size and ECG

This table presents results for conditional double sorts on size and ECG. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECG. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECG and short stocks with low ECG. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ECG Quintiles					ECG Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.22 [0.50]	0.43 [1.20]	0.64 [2.18]	0.87 [3.27]	1.05 [3.39]	0.83 [3.00]	1.07 [3.97]	0.87 [3.99]	0.83 [3.76]	0.41 [2.13]	0.39 [2.00]
	(2)	0.34 [0.87]	0.54 [1.78]	0.77 [2.91]	0.95 [3.64]	1.02 [3.63]	0.68 [3.12]	0.93 [4.42]	0.76 [4.63]	0.74 [4.44]	0.32 [2.37]	0.31 [2.31]
	(3)	0.34 [0.99]	0.65 [2.33]	0.77 [3.10]	0.94 [3.91]	1.04 [3.91]	0.70 [3.77]	0.89 [4.93]	0.74 [5.19]	0.69 [4.74]	0.31 [2.61]	0.28 [2.34]
	(4)	0.65 [2.12]	0.67 [2.68]	0.83 [3.66]	0.87 [3.81]	0.94 [3.84]	0.28 [1.84]	0.46 [3.09]	0.36 [2.80]	0.32 [2.43]	0.01 [0.13]	-0.01 [-0.08]
	(5)	0.44 [1.75]	0.63 [2.80]	0.64 [3.16]	0.94 [4.62]	0.88 [4.22]	0.44 [3.02]	0.59 [4.14]	0.61 [4.46]	0.48 [3.58]	0.20 [1.69]	0.11 [0.97]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ECG Quintiles					ECG Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	439	452	456	458	448	39	46	54	63	60	
	(2)	132	132	134	133	133	88	91	94	97	96	
	(3)	90	90	90	90	90	152	158	161	164	165	
	(4)	74	75	74	75	74	325	330	345	347	351	
(5)	66	66	66	66	66	2068	1881	2288	3264	3645		

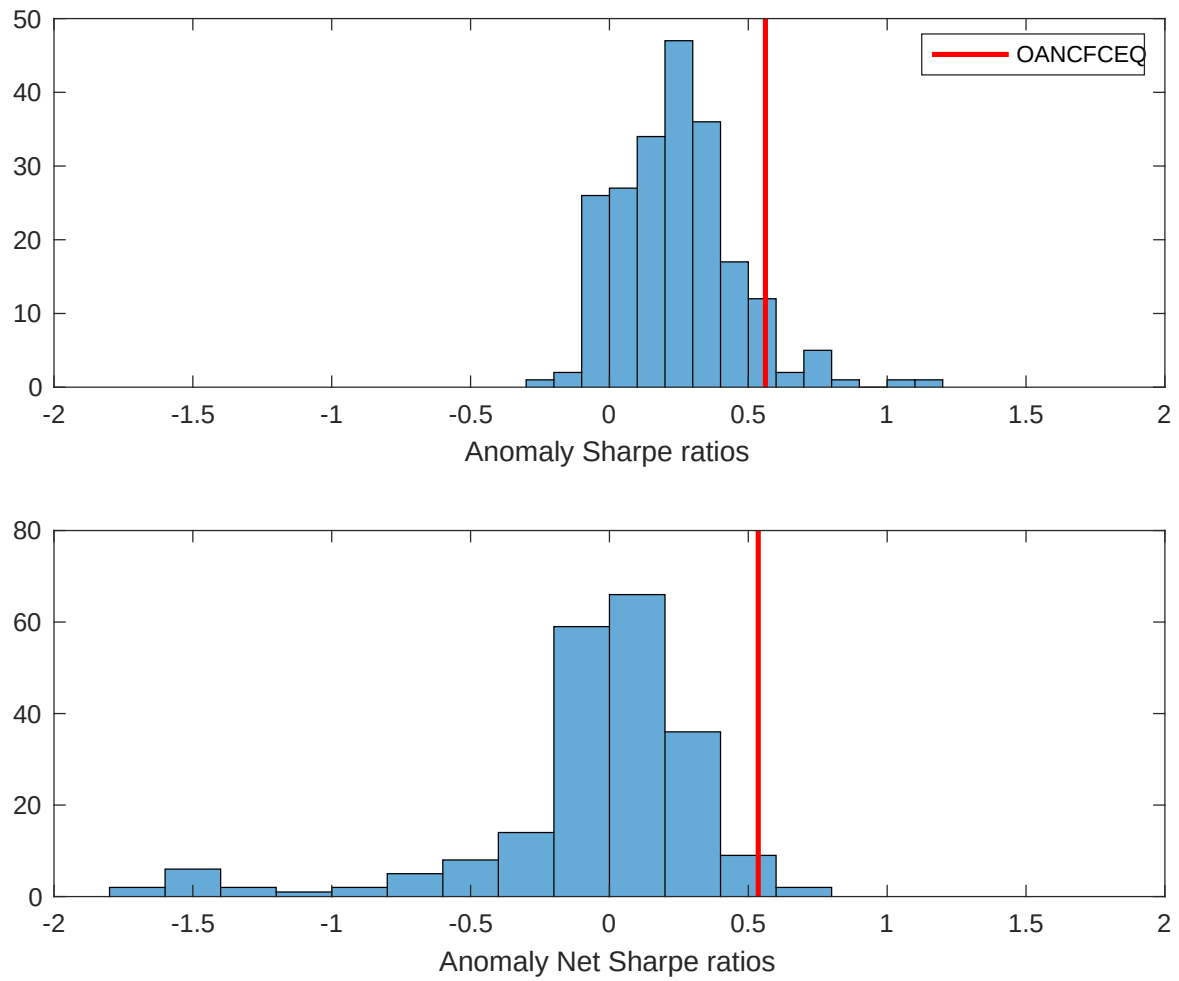


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECG with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

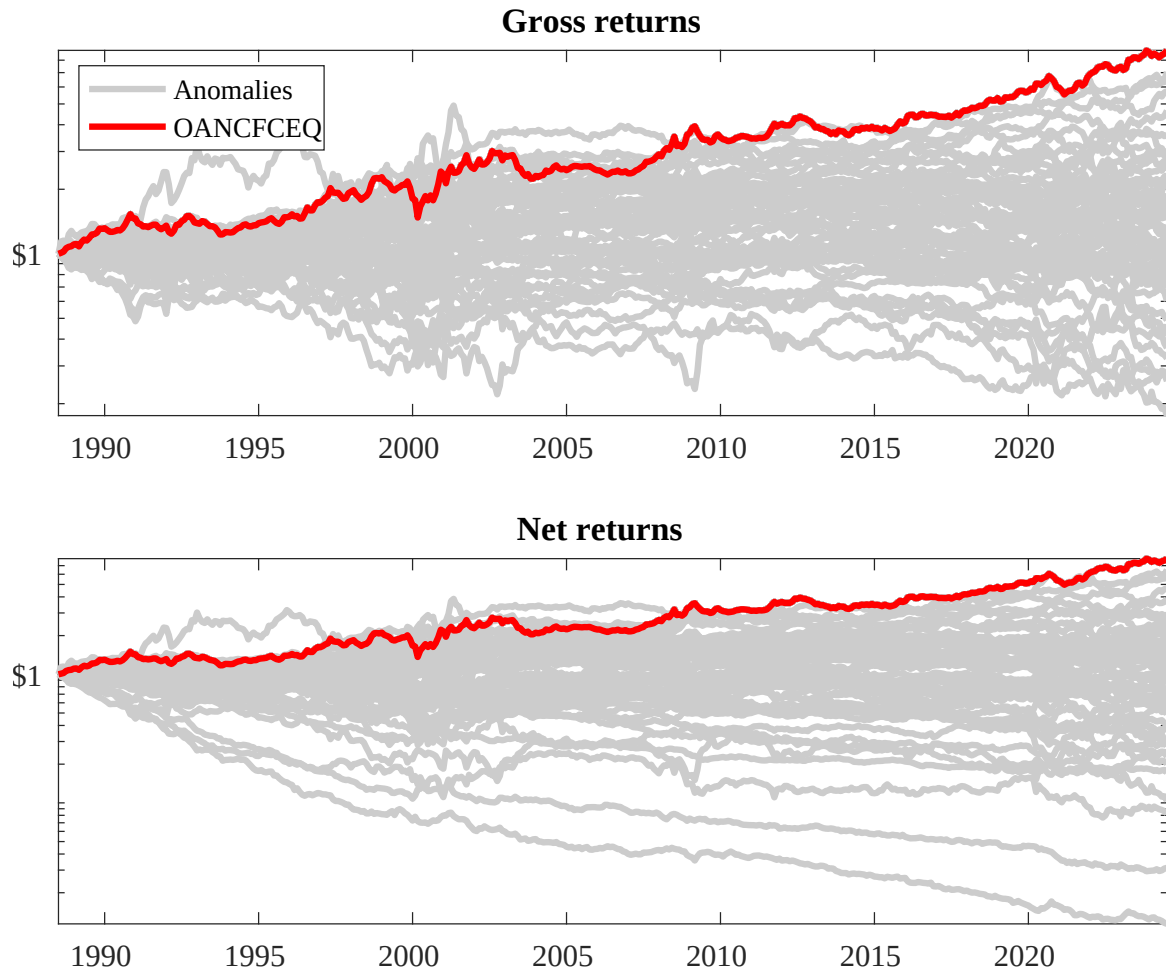
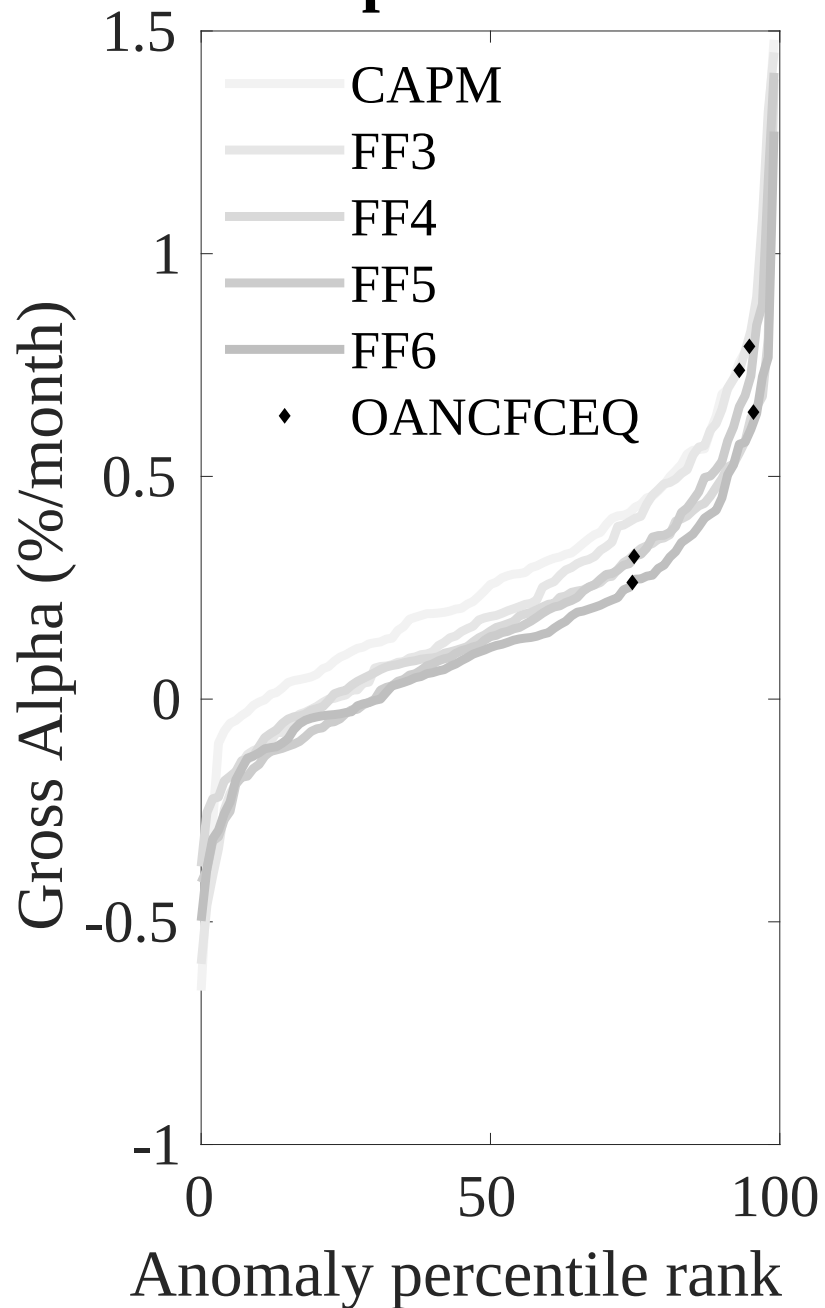


Figure 3: Dollar invested.

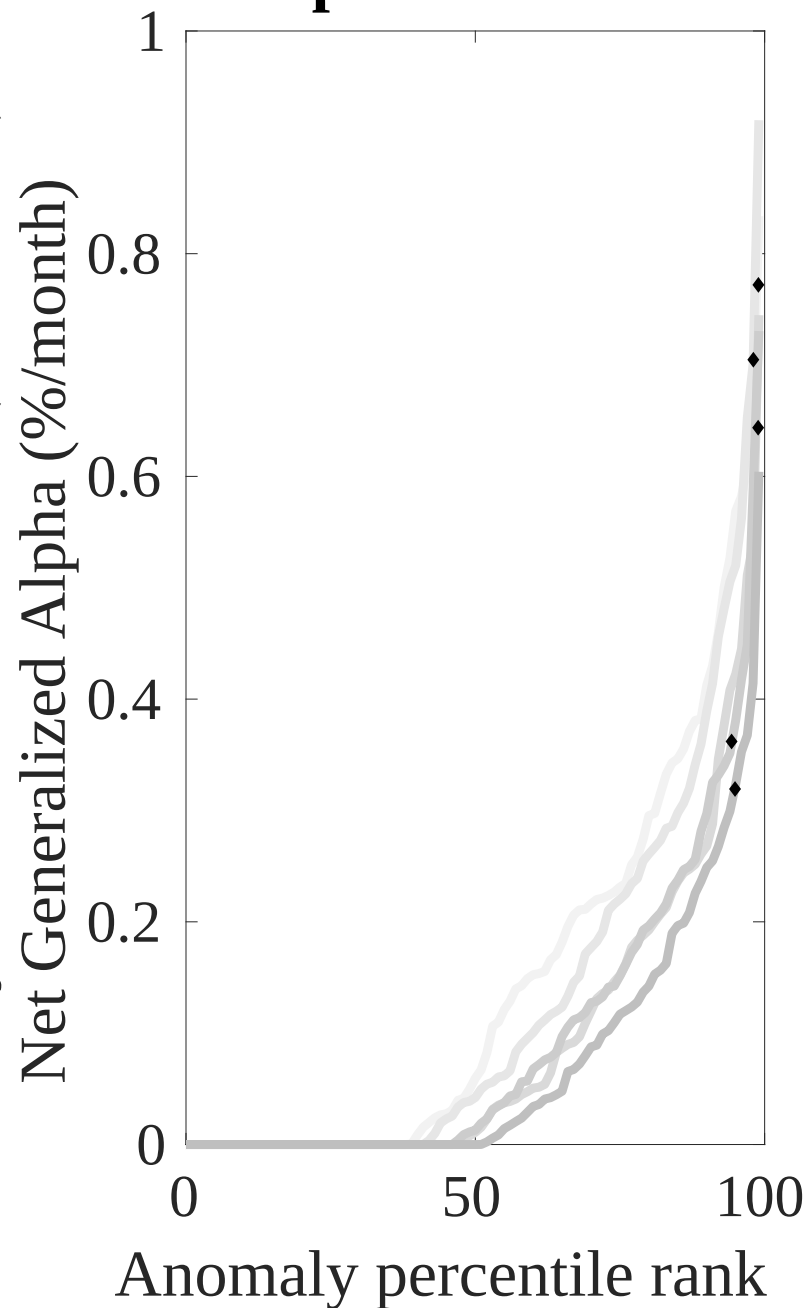
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECG trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

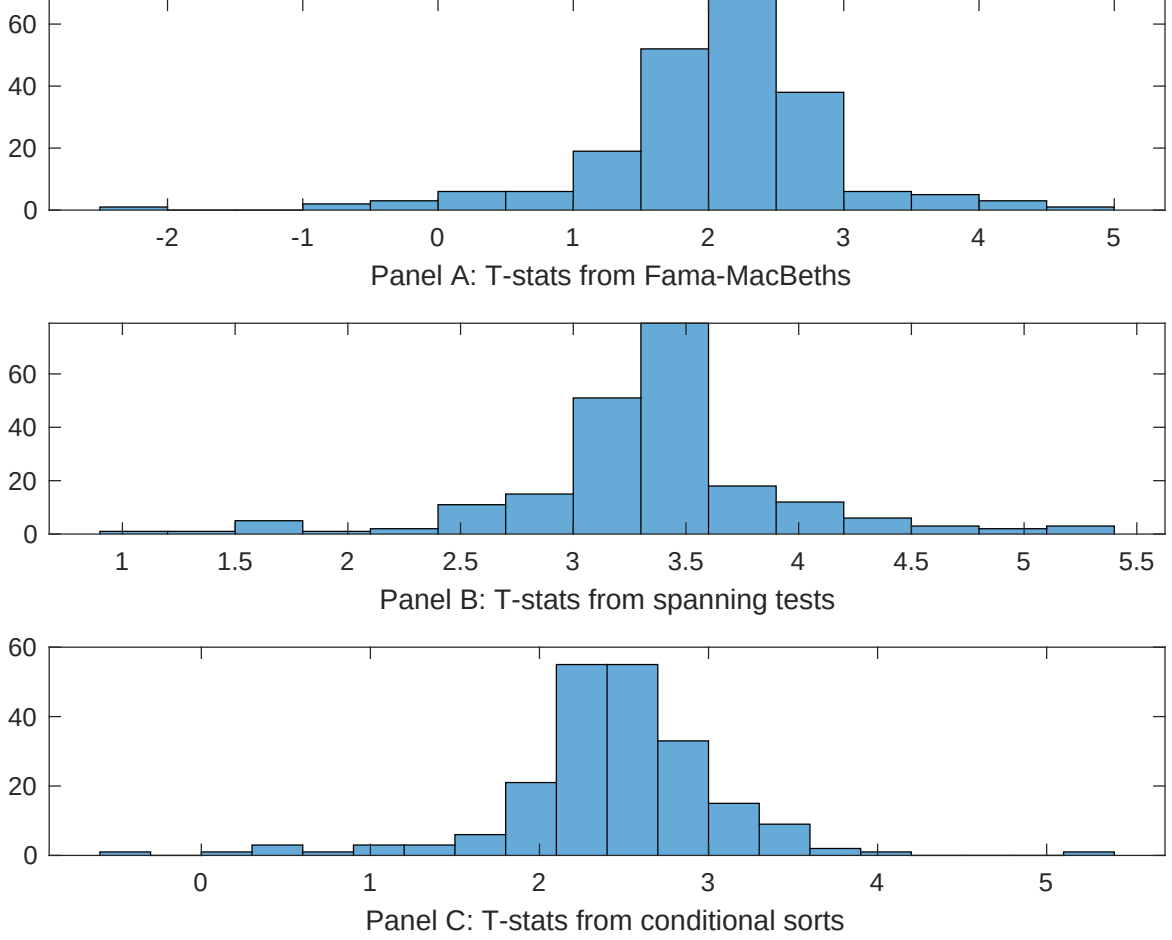


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECG conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECG} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECG} ECG_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECG,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECG. Stocks are finally grouped into five ECG portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECG trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ECG. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECG} ECG_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90 [3.03]	0.10 [3.48]	0.87 [2.54]	0.12 [4.26]	0.11 [4.76]	0.11 [4.00]	0.11 [5.09]
ECG	0.18 [2.82]	0.23 [2.85]	0.17 [2.68]	0.87 [0.11]	0.20 [3.77]	0.52 [0.91]	0.19 [2.83]
Anomaly 1	0.22 [2.44]						-0.45 [-0.56]
Anomaly 2		-0.15 [-0.28]					-0.14 [-2.59]
Anomaly 3			0.51 [0.75]				-0.14 [-0.49]
Anomaly 4				0.17 [5.76]			0.11 [4.38]
Anomaly 5					0.74 [1.22]		0.60 [0.75]
Anomaly 6						0.42 [2.74]	0.36 [3.20]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	0	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECG trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECG} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.23 [2.73]	0.25 [2.78]	0.27 [2.89]	0.22 [2.46]	0.27 [2.90]	0.25 [2.76]	0.21 [2.53]
Anomaly 1	35.66 [8.80]						36.31 [7.54]
Anomaly 2		22.45 [4.09]					-2.95 [-0.48]
Anomaly 3			-1.89 [-0.54]				-17.06 [-4.19]
Anomaly 4				22.03 [5.01]			12.32 [2.58]
Anomaly 5					4.42 [1.50]		5.05 [1.54]
Anomaly 6						6.25 [1.53]	4.08 [0.99]
mkt	-2.55 [-1.18]	-3.38 [-1.42]	-6.89 [-2.84]	-3.29 [-1.42]	-4.63 [-1.78]	-5.58 [-2.36]	-2.20 [-0.92]
smb	-29.46 [-9.24]	-30.87 [-8.39]	-39.23 [-9.99]	-30.75 [-8.75]	-34.33 [-8.30]	-36.83 [-10.86]	-31.40 [-7.97]
hml	-14.44 [-3.86]	-17.55 [-4.40]	-19.17 [-4.72]	-17.17 [-4.35]	-20.64 [-5.05]	-17.40 [-4.11]	-10.61 [-2.62]
rmw	44.29 [8.44]	56.33 [8.97]	78.00 [13.71]	62.69 [13.01]	71.23 [13.71]	70.23 [12.60]	48.72 [7.43]
cma	23.16 [4.25]	33.83 [5.87]	31.77 [5.34]	17.13 [2.70]	29.53 [4.97]	30.83 [5.28]	17.79 [2.90]
umd	5.19 [2.67]	8.29 [4.10]	9.97 [4.58]	8.83 [4.45]	8.55 [3.99]	7.96 [3.48]	6.38 [2.92]
# months	428	432	428	432	428	428	428
$\bar{R}^2(\%)$	77	73	73	74	73	73	78

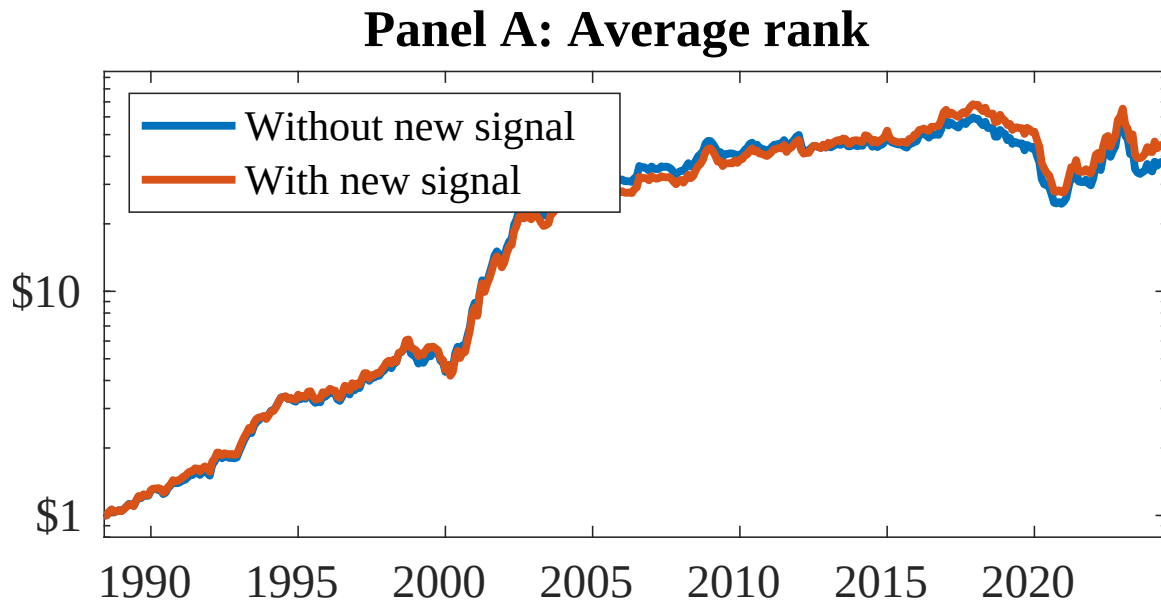


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ECG. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Ball, R., Gerakos, J., Linnainmaa, J. T., and Nikolaev, V. (2020). Earnings, retained earnings, and book-to-market in the cross section of expected returns. *Journal of Financial Economics*, 135(1):231–254.
- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2006). Profitability, investment and average returns. *Journal of Financial Economics*, 82(3):491–518.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Hirshleifer, D., Hou, K., Teoh, S. H., and Zhang, Y. (2004). Do investors overvalue firms with bloated balance sheets? *Journal of Accounting and Economics*, 38:297–331.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016a). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2016b). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Journal of Finance*, 78(3):1703–1751.